Intelligent Property Management Workflow with OCR-Enabled Automated Billing Classification and Tenant Notification

Members:

Casas, William Jake D.

Estayani, Khristian Maenard D.

Magtapat, Jullian Matthew B.

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CHAPTER I

THE PROBLEM AND ITS BACKGROUND

This chapter presents the background and related literature of the study entitled “Intelligent Property Management Workflow with OCR-Enabled Automated Billing Classification and Tenant Notification.” It also presents the significance of the study, the statement of the problem, and the scope and limitations of the system.

1.1 Introduction

Property management involves maintaining tenant records, managing utility bills, computing charges, monitoring payment status, and providing billing information to tenants. These activities require the organized handling of property records, tenant information, billing documents, and other administrative records. Cammarata (2023) discusses the importance of records and information management in real-estate operations, while Teo et al. (2024) examine property-management functions and services. When property managers handle these activities through separate or manual methods, organizing information and maintaining consistent work processes can become more difficult.

To support these activities, digital approaches provide a way to centralize property information and organize regular management tasks. Property Technology (PropTech) refers to the use of digital technologies in real-estate and property-management

activities. Siniak et al. (2020) discuss the role of PropTech in changing real-estate operations and services, while Oyetunji et al. (2025) examine digital solutions for streamlining property-management operations. Ding and Sun (2024) review technology innovations in housing management, while Al-haimi et al. (2025) discuss the benefits and challenges of digital transformation in the real-estate industry. Overall, this literature shows how digital technologies support property and housing-management activities and provides a basis for using centralized systems to organize property information and management processes.

One important area in property management is billing. Utility billing requires information from billing documents before charges can be processed, including amounts, dates, billing periods, providers, and account or reference identifiers. Optical Character Recognition (OCR) provides a way to extract text and relevant information from supported documents such as receipts and invoices (Kumar et al., 2020). Saout et al. (2024) discuss invoice-data extraction as a process involving document digitization, OCR, information extraction, and recognition of document structures. Mohan (2023) also presents a staged pipeline for extracting key-value information from commercial invoices. Overall, these studies support the use of OCR and document-processing methods to extract and organize information from billing documents for later processing.

After billing information is extracted, it still needs to be validated, classified, and processed according to defined billing conditions. Rule-based processing uses predefined conditions and logical rules to support structured operational decisions and

information handling. Baralis et al. (2022) discuss business-rule engines that represent and execute operational decisions in a structured way. In relation to process automation, Geyer-Klingeberg et al. (2024) present a rule-based method that analyzes decision logic and classifies business processes according to their suitability for robotic process automation. These ideas are relevant to a billing workflow in which validation, classification, calculation, and exception-handling conditions are defined in advance.

Based on these ideas, the proposed system connects property management, OCR-based extraction of supported utility-bill documents, validation, rule-based billing classification, billing computation, exception handling, and Web Push Notifications within one web-based workflow. In this study, the term “intelligent” refers to the combined use of OCR-enabled document text extraction and predefined, explicit, deterministic, and traceable rules for billing-workflow support. The OCR component may use machine-learning-based text-recognition technology to identify and extract available text from supported documents; however, OCR output is not treated as automatically validated billing data. Validation, duplicate checking, billing classification, billing computation, exception routing, and payment confirmation are handled through defined rules and Manager-controlled actions. The study focuses on organizing rental and utility billing activities within a centralized system while applying defined rules to billing and exception-handling processes.

1.2 Review of Related Literature

Digital Property Management and PropTech

Digital property management increasingly relies on technology to organize property details, management activities, billing data, and related services. Several studies have examined this shift from different perspectives: Teo et al. (2024) discuss technology and digitalization in property-management functions and service delivery, while Ding and Sun (2024) review technology innovations in housing management. In this context, PropTech refers to the use of digital technologies in the real estate and property-management sector. Siniak et al. (2020) examine the impact of PropTech on real estate industry growth, Saputra et al. (2023) review digital transformation trends in the property-management industry, and Al-haimi et al. (2025) synthesize the technologies, benefits, and challenges associated with digital transformation in real estate. In addition, Cammarata (2023) discusses the importance of records and information management in managing risks during real estate changes. Overall, this literature supports the relevance of centralized digital tools for property information and administrative processes. Building on this literature, the present study proposes a centralized web-based workflow that organizes property, unit, tenant, utility, billing, payment, exception, and receipt information for selected boarding houses and residential apartments.

OCR and Billing Document Extraction

Within property-management operations, billing is one area where digital processing can support routine administrative work. Billing documents contain information such as amounts, dates, billing periods, providers, and account or reference identifiers. Kumar et al. (2020) examine the extraction of information from bill receipts using Optical Character Recognition (OCR), while Saout et al. (2024) provide an overview of invoice-data extraction involving document digitization, OCR, information extraction, and document-structure recognition. Similarly, Mohan (2023) presents an end-to-end pipeline for custom key-value extraction from commercial invoices. However, extraction is only the first stage of billing processing. In the proposed system, OCR is used to process supported standard digital PDFs and clear images, after which the extracted information undergoes validation before billing processing. Records that do not meet the required validation conditions are sent to the Exception Queue for Manager review. Thus, OCR is treated as a method for extracting information from billing documents rather than as a guarantee that every extracted value is accurate. The OCR component may employ machine-learning-based text-recognition technology to convert visible characters and document text into machine-readable output. However, the proposed system does not use OCR output or machine-learning recognition to determine billing validity, classify payment status, compute charges, or make property-management decisions.

Rule-Based Billing Processing

Once billing information has been extracted, it must be checked and processed according to defined conditions. Rule-based processing uses explicit conditions and decision rules to classify, validate, and route information. Wu et al. (2022) demonstrate the use of rule-based information extraction to organize domain-specific information, while Baralis et al. (2022) discuss business-rule engines for representing and executing operational decisions in a structured manner. Extending this idea to process automation, Geyer-Klingeberg et al. (2024) present a rule-based method that analyzes decision logic and classifies business processes according to their suitability for robotic process automation, while Kim et al. (2011) examine proactive exception handling in business processes. These studies support the use of predefined and traceable conditions to determine whether a record proceeds through normal processing or requires additional review. Applying these principles, the proposed system uses defined rules for billing validation, provider and account/reference mapping, duplicate detection, and identifying individual or shared billing arrangements. Records that do not meet the defined conditions are sent to the Exception Queue for Manager review. These rules, as well as the separation between Tenant Reported Paid and Manager-confirmed payment status, are system requirements rather than findings established by the cited studies. Digital receipts are created only after payment confirmation.

Shared Utility Billing and Cost Allocation

Shared utility billing presents a related issue because a single charge may need to be allocated among several occupants. Levinson and Niemann (2004) examine

apartment energy use when landlords pay for utilities, while Mayr and Souza (2025) discuss coordination and incentive issues associated with shared heating costs. These studies provide context for understanding cost responsibility and consumption incentives in shared-utility arrangements. Nevertheless, they do not establish one universal allocation formula for rental properties or determine how the proposed system should calculate individual tenant charges. For this reason, the proposed system treats the allocation method as a defined system requirement rather than as a formula established by the reviewed literature. For shared occupancy, the Manager configures the predefined allocation rule. The billing requirements also include reconciliation controls so that allocated charges equal the source billing amount. The specific allocation formula, rounding procedure, and reconciliation method are system-design decisions described and tested in the succeeding chapters.

Web Push Notifications and Secure Notification Delivery

After a billing record is finalized, tenants need to be informed that their billing information is available. The World Wide Web Consortium (2024a) describes the Push API, including push services and user subscriptions. The World Wide Web Consortium (2024b) also explains the role of service workers in handling web events in the background when they are included in the system implementation. Because Web Push involves browser subscriptions and authorization, security must also be considered. Carboneri et al. (2023) examine Web Push implementations in the wild and identify security concerns involving push subscriptions and authorization. In the proposed

system, Web Push Notifications serve as billing alerts that direct tenants to the authenticated web portal, where complete billing information is available. However, Web Push does not confirm payment, prove that payment occurred, or show whether the tenant read the notification. Tenant payment reporting therefore remains separate from Manager confirmation.

Iterative Software Development

Because the proposed system contains several connected functions, its development requires an organized and adaptable process. Iterative software development divides development into repeated cycles of building, reviewing, receiving feedback, and improving. Larman and Basili (2003) explain the development of iterative and incremental approaches, in which software is developed progressively through repeated cycles. Inayat et al. (2015) emphasize stakeholder involvement, communication, and requirements refinement in iterative requirements engineering, while Ramesh et al. (2021) identify challenges involving changing requirements, stakeholder participation, documentation, and communication in Agile software development. These studies support the use of Iterative Software Development Methodology in the present study. Specifically, the methodology organizes the proposed system into repeated cycles of requirements review, development, testing, feedback, and refinement. Although the literature supports this approach, the actual quality and performance of the proposed system are determined through implementation, testing, user feedback, and evaluation.

Software Quality Evaluation and Security

To evaluate the proposed system systematically, the study requires defined software-quality criteria. ISO/IEC 25010:2023 provides a product-quality model for systems and software quality requirements and evaluation (International Organization for Standardization, 2023). Guided by this model, the study evaluates five quality characteristics: Functional Suitability, Performance Efficiency, Interaction Capability, Reliability, and Security. These characteristics provide a basis for examining whether the system performs its intended functions, performs efficiently, supports user interaction, operates consistently, and protects information. Security is particularly important because the system handles tenant and billing information. Republic Act No. 10173, or the Data Privacy Act of 2012, requires organizations that process personal information to use reasonable and appropriate measures to protect it (Republic of the Philippines, 2012). The National Privacy Commission (2016) further discusses safeguards for the confidentiality, integrity, and availability of personal data, while Zafar et al. (2022) review authorization and access-control strategies that restrict system functions and information according to user permissions. In the proposed system, role-based access separates Manager and Tenant functions. These sources guide the study's quality and security evaluation, but they do not prove that the system already meets the selected quality characteristics or privacy requirements. The Security evaluation is limited to the defined system functions, access controls, and security-related criteria included in the study's evaluation procedures. It does not constitute a legal audit, certification, or claim of complete compliance with the Philippine Data Privacy Act.

1.3 Significance of the Study

The study develops and evaluates a web-based property-management workflow that uses OCR-based information extraction, rule-based billing classification, and Web Push Notifications. It primarily benefits property managers and tenants and may serve as a reference for future researchers and system developers.

In the global context, the study contributes an applied example of PropTech and digital property management by showing how existing technologies can be combined into one workflow for residential property operations. It also provides a local example of how digital property-management concepts can be applied in a smaller property-management setting. The study does not claim that the proposed workflow represents all property-management environments; rather, its significance lies in demonstrating how selected technologies may be integrated within the defined scope of the study.

In the economic context, the system is designed to reduce repetitive manual activities involved in receiving utility bills, recording billing information, calculating tenant charges, and communicating billing updates. Centralized records and predefined billing rules can also help property managers follow a more consistent billing process. However, actual reductions in operating costs, processing time, or late payments depend on the results of system testing and evaluation.

In the environmental context, the system uses electronic utility bills, digital billing records, and electronic notifications. This can reduce the need for paper-based

records during the billing process. The study does not claim a specific reduction in paper use because this is not directly measured.

In the societal context, the system provides property managers and tenants with organized billing information. Itemized billing records, defined billing rules, and automated notifications can help make tenant dues easier to understand and communicate. This can support clearer billing transactions between property managers and tenants. However, the study does not claim that the system will eliminate billing misunderstandings or disputes because those outcomes are not directly measured.

1.4 Statement of the Problem

This study develops and evaluates an Intelligent Property Management Workflow with OCR-Enabled Automated Billing Classification and Tenant Notification for selected boarding houses and residential apartments in Pasig City.

Specifically, the study seeks to answer the following questions:

1. What development stages are undertaken in developing the proposed system using the Iterative Software Development Methodology?
2. How does the proposed system integrate OCR, rule-based billing processing, exception handling, payment management, digital receipts, and Web Push Notifications into one property-management workflow?

3. How does the developed system perform when evaluated using the selected ISO/IEC 25010:2023 quality characteristics in terms of:

    a. Functional Suitability;

    b. Performance Efficiency;

    c. Interaction Capability;

    d. Reliability; and

    e. Security?

4. What implementation guide and user manual can be prepared for the developed system?

1.5 Scope and Limitations of the Study

The study develops and evaluates a web-based property-management workflow entitled Intelligent Property Management Workflow with OCR-Enabled Automated Billing Classification and Tenant Notification. It aims to provide a centralized workflow for organizing property, unit, tenant, utility, rent, billing, payment, exception, receipt, and notification records. The system integrates electronic utility-bill intake, OCR-based document-information extraction, validation, deterministic rule-based billing classification, Exception Queue handling, Manager payment confirmation, digital receipt generation, and Web Push Notifications. In this study, the term “intelligent” refers to the combined

use of OCR-enabled document text extraction and predefined, explicit, deterministic, and traceable rules for billing-workflow support. The OCR component may use machine-learning-based text-recognition technology to extract available billing information from supported documents. However, the system does not use Artificial Intelligence, Machine Learning, Deep Learning, Predictive Analytics, or Generative AI to make autonomous decisions involving record validation, billing classification, charge computation, cost allocation, payment confirmation, receipt authorization, or property management. The OCR component is limited to document text recognition and extraction, while the remaining processing follows predefined rules and Manager-controlled actions.

The study covers selected boarding houses and residential apartments in Pasig City during Academic Year 2026–2027. The billing categories are limited to rent, electricity, and water. Rent will be generated once per monthly lease cycle using the configured lease-anniversary date. The monthly rent will be fixed and non-prorated. When the anniversary day does not occur in a particular month, the system will use the last calendar day of that month while retaining the original anniversary day for succeeding months. For example, a lease beginning on January 31 will use February 28, or February 29 in a leap year, followed by March 31. The rent-generation date and rent due date will be the same under the study's billing policy, and the corresponding coverage period will be recorded in the billing entry. The first billing record will contain the applicable first-cycle rent and a separate line item for one-month advance rent. The advance rent will remain within the rent category and will represent credit for the final

applicable monthly rent cycle of the lease. It will not be treated as a security deposit or charged again in subsequent cycles. For example, an individual monthly rent of PHP 5,000.00 will produce a first billing record containing PHP 5,000.00 for current rent and PHP 5,000.00 for advance rent. Once the payment has been confirmed, the advance portion will be recorded as available rent credit. For shared occupancy, each tenant's first-cycle share and advance-rent share will be shown separately. The system will support Individual Billing Arrangement and Shared-Occupancy Billing Arrangement. Under individual billing, the applicable charge will be assigned to the tenant associated with the configured lease or utility account. Under shared occupancy, the source rent or utility charge will be divided equally among the eligible tenants in the configured billing group. The Manager will identify the property or unit covered by the source charge and maintain the tenant assignments used for allocation. The study will use equal splitting only; it will not estimate consumption or use meter readings to determine individual shares.

The system receives electronic utility bills through a designated property-management email account and uses OCR to extract available billing information from standard, digitally generated PDF files and sufficiently clear image files. The target fields include the amount due, due date, billing period, utility provider, and account or reference identifier. OCR is limited to document-information extraction and does not determine billing classification, confirm payment, or replace validation. Extracted billing information is checked for required fields, provider and account/reference mapping, billing period, duplicate status, and other defined validation



conditions. Records that are incomplete, uncertain, duplicated, unmappable, or unable to satisfy the defined rules are sent to the Exception Queue for Manager review and correction. Corrected records may be revalidated and reprocessed.

The system processes valid records according to the defined billing and computation rules. It uses three payment statuses: Unpaid, Tenant Reported Paid, and Paid. Tenant Reported Paid indicates a tenant-reported payment and does not represent confirmed payment. Only the Manager can confirm payment and change the status to Paid. A digital receipt is generated only after Manager-confirmed payment. The system does not include a payment gateway, bank integration, automatic payment verification, independent external payment verification, or direct utility-payment verification. The Manager performs any required payment verification before confirming payment in the system.

The system provides a tenant-facing dashboard for viewing recorded rental and utility dues, billing details, payment status, and confirmed-payment receipts. Tenant alerts use Web Push Notifications and direct tenants to the authenticated web portal. The system does not use SMS or email as tenant-notification channels. Web Push Notifications provide proactive billing updates and due-date-related alerts, including a configured reminder three days before the due date. They do not confirm payment, prove that payment occurred, prove that a tenant read the notification, or replace the billing record. Current billing information, payment status, and confirmed-payment receipts remain accessible through the tenant's authenticated web portal regardless of push

permission, delivery, or device support. Notification delivery behavior is described and evaluated according to the implemented notification mechanism.

The system includes Admin, Manager, and Tenant roles. The Admin role is limited to system maintenance. The Manager manages properties, units, tenants, utility mappings, billing rules, billing records, exceptions, corrections, payment confirmation, and receipts. The Tenant views billing information and payment status, reports payment, and receives permitted Web Push Notifications. Role-based access restricts functions and information according to the assigned role.

The system uses a centralized database for property, unit, tenant, utility-mapping, rent-configuration, billing, payment-status, exception, correction, receipt, and applicable notification records.

The system is evaluated using five selected ISO/IEC 25010:2023 quality characteristics: Functional Suitability, Performance Efficiency, Interaction Capability, Reliability, and Security. The evaluation is limited to these selected characteristics and does not represent an assessment of every characteristic in the ISO/IEC 25010:2023 product-quality model. The study evaluates the system according to its defined functions and approved testing procedures and does not claim to measure long-term changes in property revenue, tenant behavior, tenant payment compliance, business profitability, or overall operational performance unless these outcomes are specifically included in the evaluation.

Within its defined boundaries, the study does not include a mobile application; installation or physical reading of Internet of Things smart meters, electrical submeters, or water submeters; Internet or Wi-Fi billing; predictive profit analytics; predictive billing; Generative AI; autonomous AI-based billing decisions; blockchain technology; automated utility disconnection; payment gateways; bank integration; independent external payment verification; maintenance-ticket management; legal contract generation; or automated eviction processing. Although the OCR component may use machine-learning-based text-recognition technology for document text extraction, no machine-learning model is trained or customized by the researchers, and no machine-learning model is used to make autonomous billing, payment, allocation, validation, or property-management decisions. The OCR component does not cover handwritten bills, heavily damaged documents, severely distorted documents, or low-resolution photographs. The study also does not claim that the system independently verifies real-world payments, guarantees notification delivery or reading, establishes a universally fair utility-allocation formula, or provides complete legal certification under the Philippine Data Privacy Act.

CHAPTER II

METHODOLOGY

This chapter presents the methodology and procedures that will guide the development of the proposed Intelligent Property Management Workflow with OCR-Enabled Automated Billing Classification and Tenant Notification. It describes the project requirements, system design, conceptual operation, interface mockups, and Iterative Software Development Methodology. It also presents the testing and evaluation procedures that will be used to examine the developed system within the approved scope of the study.

2.0 Project Overview

Project Information

The project, entitled Intelligent Property Management Workflow with OCR-Enabled Automated Billing Classification and Tenant Notification, is a web-based system for selected boarding houses and residential apartments in Pasig City. It supports property, unit, tenant, rent, electricity, water, utility-mapping, billing, payment-status, exception, receipt, and applicable notification records. The system provides an

integrated workflow for rent and utility billing. It receives electronic utility bills through the designated property-management email account, retrieves supported attachments, extracts available billing information through OCR, validates and processes records using predefined rules, computes and finalizes billing records, manages exceptions, records payment status, generates receipts after Manager-confirmed payment, and provides permitted Web Push billing and due-date alerts.

In this study, intelligent refers to OCR-enabled document text extraction combined with predefined, explicit, deterministic, and traceable rules. The OCR component may use machine-learning-based text recognition to extract available document information, but it does not make autonomous billing, validation, allocation, payment, or property-management decisions. These functions use defined rules and Manager-controlled actions.

Requirements Definition

The proposed system is designed to meet the following business, user, and security requirements.

A. Business Requirements

The proposed system shall:

1. Receive electronic utility bills through the designated property-management email account and retrieve supported billing attachments.

2. Use OCR to extract available billing information from supported standard, digitally generated PDF files and sufficiently clear image files, including the amount due, due date, billing period, utility provider, and account or reference identifier.

3. Validate extracted billing information using predefined conditions for required fields, provider and account/reference mapping, billing period, duplicate status, and other defined conditions. Utility records shall be mapped to the appropriate property, unit, utility account, or reference identifier and provider through Manager-configured mappings.

4. Apply predefined, explicit, deterministic, and traceable rules for billing validation, provider and account/reference mapping, duplicate detection, billing classification, individual or shared billing identification, and billing computation according to configured rules.

5. Process billing categories limited to rent, electricity, and water. The system shall support Individual Billing Arrangement and Shared-Occupancy Billing Arrangement.

6. Generate fixed, non-prorated rent once per monthly lease cycle using the lease-anniversary date. Use the last calendar day for months without the anniversary day and retain the original day for later months. Record one-month advance rent as a separate rent line in the first billing record. After payment

confirmation, retain the advance as credit for the lease's final applicable rent cycle and prevent repeated billing or repeated use of the credit.

7. Allocate shared rent and applicable utility charges equally among the eligible tenants in the Manager-configured billing group. Determine rent eligibility on the rent-generation date and utility eligibility on the final day of the source billing period. Record the eligible tenant list, compute shares in whole centavos, and distribute remaining centavos in ascending tenant-record identifier order. The allocated shares shall total the original source charge. Records without a valid allocation group or eligible tenant list shall enter the Exception Queue.

8. Process and finalize billing records when required validation, mapping, duplicate-checking, and billing conditions are satisfied. Records that do not satisfy the defined processing conditions shall be routed to the Exception Queue for Manager review and correction.

9. Allow corrected exception records to be revalidated and reprocessed or recomputed when applicable. Corrections to finalized records shall undergo Manager correction, revalidation, recomputation when applicable, and record update.

10. Store property, unit, tenant, utility-mapping, rent-configuration, billing, payment-status, exception, correction, receipt, and applicable notification information in a centralized database.

11. Maintain the payment statuses Unpaid, Tenant Reported Paid, and Paid. The system shall distinguish tenant-reported payment from Manager-confirmed payment. Only the Manager may confirm a tenant-reported payment and change the status to Paid.

12. Generate a digital receipt only after Manager-confirmed payment.

13. Provide a tenant-facing dashboard for viewing recorded rental and utility dues, billing details, payment status, and confirmed-payment receipts.

14. Trigger permitted Web Push Notifications for billing updates and due-date-related alerts, including a configured reminder three days before the due date. Web Push Notifications shall direct tenants to the authenticated web portal where complete billing information is available.

15. Maintain the distinction between routine automated processing and records requiring Manager intervention.

16. The system shall not use SMS, GSM, or tenant email notifications. The designated property-management email account shall be used only for utility-bill intake and attachment retrieval.

B. User Requirements

The proposed system includes three user roles: Admin, Manager, and Tenant.

Admin

The Admin shall be able to perform only the system-maintenance functions included in the implemented system.

Manager

The Manager shall be able to:

1. Log in to the system using valid credentials.

2. Manage property, unit, and tenant information.

3. Configure utility providers, account or reference mappings, rent information, billing arrangements, and applicable billing rules, including the shared-occupancy allocation rule.

4. View and manage billing records.

5. Review OCR-extracted billing information and validation and classification results when necessary.

6. Review records placed in the Exception Queue.

7. Correct billing or payment information when Manager intervention is required, including post-finalization corrections, and submit corrected records for revalidation and reprocessing or recomputation when applicable.

8. Confirm tenant-reported payments.

9. View payment status for applicable rent, electricity, and water billing records.

10. View confirmed-payment receipts.

Tenant The Tenant shall be able to:

1. Log in to the system using valid credentials.

2. View assigned property or unit information.

3. View applicable rent, electricity, and water billing records.

4. View billing details, including billing category, billing period, and amount due.

5. View payment status.

6. Report an applicable payment through the system.

7. View digital receipts for Manager-confirmed payments.

8. Receive permitted Web Push billing and due-date alerts that direct the Tenant to the authenticated web portal.

A tenant-reported payment shall not automatically change the payment status to Paid. The status shall remain Tenant Reported Paid until the Manager confirms the payment.

C. Security Requirements

The proposed system shall protect property, tenant, billing, payment, utility, exception, correction, receipt, and applicable notification information through the following security requirements:

1. Require users to provide valid credentials before accessing protected system functions.

2. Restrict system functions and information according to the assigned Admin, Manager, or Tenant role.

3. Limit access to property, tenant, billing, payment, utility, exception, correction, receipt, and applicable notification information to authorized users and system functions.

4. Restrict access to electronic utility-bill documents retrieved through the designated property-management email workflow to authorized users and functions.

5. Protect billing and payment information from unauthorized access or modification through authentication, role-based access control, and defined system permissions.

6. Manage authenticated user sessions according to the implemented access-control mechanism.

7. Validate user-provided information and extracted billing information before storage or processing.

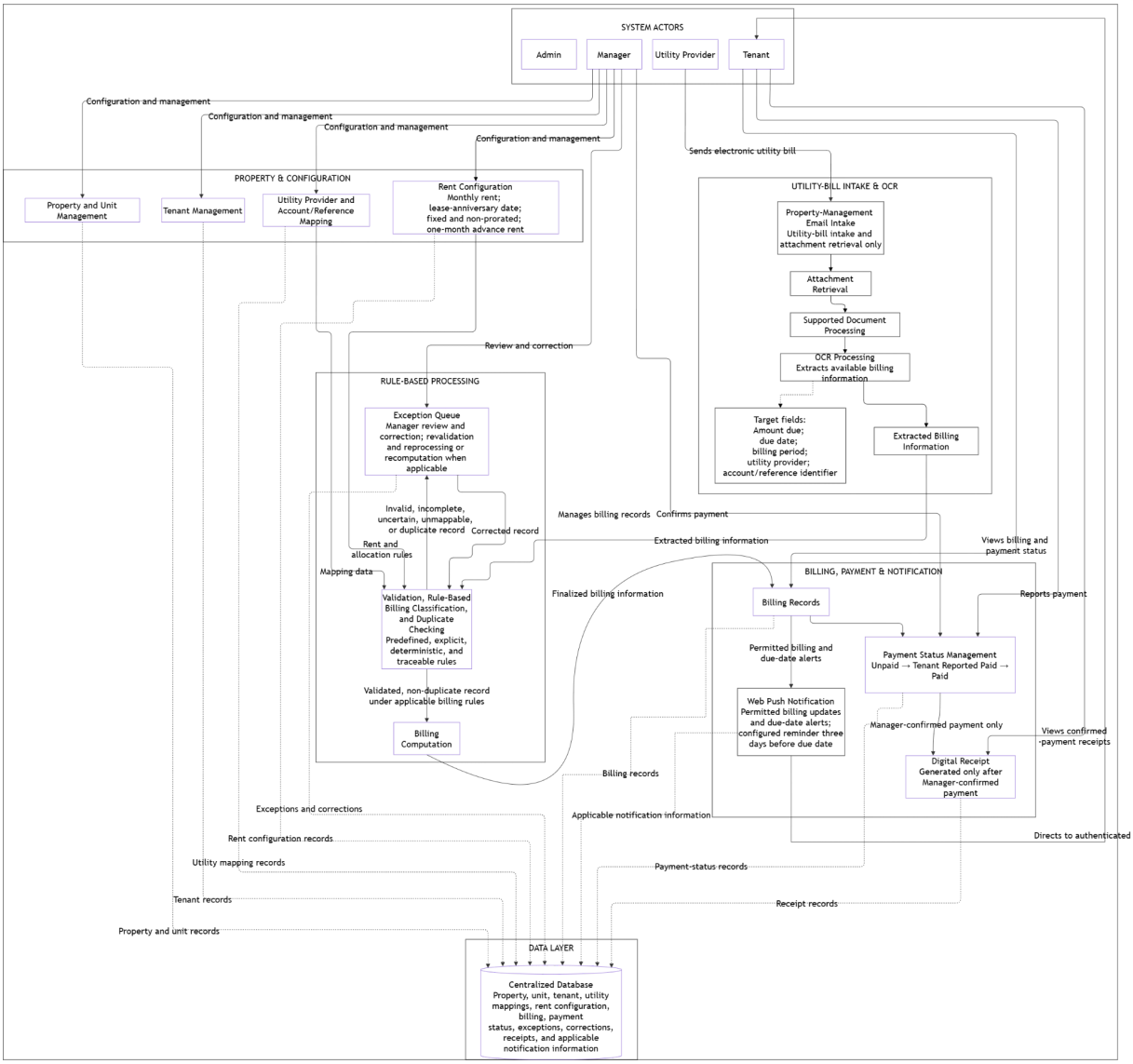
8. Route records that fail required-field validation, identification, duplicate checking, mapping, or other defined processing conditions to the Exception Queue for Manager review.

9. Store correction information in the Centralized Database as applicable.

These security requirements are limited to the implemented system functions, access controls, and security-related evaluation criteria. They do not constitute a legal audit, certification, or claim of complete compliance with the Philippine Data Privacy Act.

2.1 Concept of the Study

Architectural Framework



The architectural framework presents the structure and interaction of the proposed Intelligent Property Management Workflow with OCR-Enabled Automated Billing Classification and Tenant Notification. It shows the major components, actors, and information relationships within the proposed system. The framework organizes routine processing components and exception-handling components within the proposed

system. The Exception Queue provides a controlled point for Manager review and correction of records that do not satisfy the defined processing conditions.

The framework consists of the following major components and actors:

- Property and Unit Management
- Tenant Management
- Utility Provider and Account/Reference Mapping
- Rent Configuration
- Property-Management Email Intake
- OCR Processing
- Validation and Rule-Based Billing Classification
- Duplicate Checking
- Billing Computation
- Exception Queue
- Billing Records
- Payment Status Management
- Digital Receipt
- Web Push Notification

- Centralized Database
- Admin
- Manager
- Tenant
- Utility Provider

Property and Unit Management manages property and unit information. Tenant Management manages tenant information and its association with the applicable property or unit. The Utility Provider and Account/Reference Mapping component manages utility providers, account or reference identifiers, and their configured property or unit relationships. The Manager configures the applicable mappings.

Rent will be generated once per monthly lease cycle using the configured lease-anniversary date. The monthly rent will be fixed and non-prorated. When the anniversary day does not occur in a particular month, the system will use the last calendar day of that month while retaining the original anniversary day for succeeding months. For example, a lease beginning on January 31 will use February 28, or February 29 in a leap year, followed by March 31. The rent-generation date and rent due date will be the same under the study's billing policy, and the corresponding coverage period will be recorded in the billing entry. The first billing record will contain the applicable first-cycle rent and a separate line item for one-month advance rent. The advance rent will remain within the rent category and will represent credit for the final applicable

monthly rent cycle of the lease. It will not be treated as a security deposit or charged again in subsequent cycles. For example, an individual monthly rent of PHP 5,000.00 will produce a first billing record containing PHP 5,000.00 for current rent and PHP 5,000.00 for advance rent. Once the payment has been confirmed, the advance portion will be recorded as available rent credit. For shared occupancy, each tenant's first-cycle share and advance-rent share will be shown separately. When the Manager records the lease's final applicable cycle, the confirmed advance-rent credit will be applied to that cycle's rent charge. The billing record will show the charge, the credit applied, and any remaining amount due. Applying previously confirmed credit will not be treated as a new cash payment or generate another receipt for the same money. The system will retain the original payment and receipt reference. Advance credit will not be applied beyond its available confirmed amount. Any discrepancy requiring correction will be reviewed by the Manager through the existing correction process. The system will check whether rent has already been generated for the same lease and monthly cycle before creating another charge. The first-billing advance line will also be checked to prevent repeated creation. New rent cycles will not be generated after the recorded lease end date, while existing unpaid charges will remain available for review. Changes to tenant assignments or shared-rent membership will follow the allocation policy and will not automatically replace finalized billing entries.

Property-Management Email Intake receives supported electronic utility-bill attachments through the designated property-management email account. The email account is used only for utility-bill intake and attachment retrieval and is not used as a

tenant-notification channel. The OCR component will process supported electronic utility-bill attachments received through the designated property-management email account. Supported inputs will consist of digitally generated PDF documents and sufficiently clear image documents using the provider layouts selected for the study. The system will retain a reference to the source attachment and extract the amount due, due date, billing period, utility provider, and account or reference identifier. It will not infer payment status or determine tenant charges from the document alone. The processing procedure will use an existing OCR engine for document text recognition without training or customizing a machine-learning model. Recognized text will be examined through predefined field-extraction rules appropriate to the supported layouts. The extracted values will then be submitted to the validation and mapping procedure. OCR output will not be treated as confirmed billing information solely because text was recognized. A record with absent, unreadable, or conflicting required values will enter the Exception Queue for comparison with the source document and Manager correction. Before final testing, the researchers will document the supported provider layouts, accepted file types, and file-processing limits. Unsupported or unreadable attachments will be retained as processing exceptions and will not produce finalized charges. The evaluation will use supported document samples with manually checked field values, as described in the testing procedure. Results will be reported within those document boundaries and will not be presented as accuracy for every utility provider or document format.

After extraction, the system will check the required billing fields before computing tenant charges. Amounts will be converted to the declared currency format, dates will be



checked for validity, and account or reference identifiers will be treated as text so that leading zeros are retained. The billing period must identify a valid beginning and ending date. Missing, conflicting, or unreadable required values will prevent automatic finalization and will cause the record to enter the Exception Queue. The utility provider and account identifier will be compared with the mappings configured by the Manager. A valid mapping will identify the utility category and the property or unit covered by the bill. The individual or shared billing arrangement will be obtained from the configuration associated with that target. Provider aliases may be normalized through an explicit configured list, but the system will not guess a target when no unique mapping is available. Rent will follow the saved rent configuration and will not be classified through OCR. Duplicate checking will compare both the received attachment and the extracted billing information with existing records. An identical attachment, or a record with the same provider, account, and billing period, will be held for Manager review rather than creating another charge. If the provider has issued a revised bill for the same period, the Manager will review it through the correction process before any existing charge is updated. Reprocessing an already accepted source record will not create a second allocation or billing entry. For each processed record, the system will retain the source reference, normalized field values, applied rule identifiers, and processing result. These records will allow the Manager to identify why a bill was accepted, classified, or placed in the Exception Queue. A corrected record will undergo the same required checks before it can return to computation and finalization.

Table 2.1: Validation and classification rules

Rule ID	Rule Name	Description	Action/Result
V01	Required information	Amount, due date, billing period, provider, or account identifier is missing or conflicting	Exception Queue; no finalization
V02	Valid values	Amount or date cannot be interpreted, amount is negative, or period begins after it ends	Exception Queue; Manager review
V03	Unique mapping	Provider/account identifies exactly one configured target and category	Continue; otherwise Exception Queue
V04	Duplicate check	Identical attachment or same provider/account/period is already recorded	Hold for duplicate or revised-bill review
C01	Utility category	Configured provider/account mapping identifies electricity or water	Assign mapped category
C02	Billing arrangement	Target has a valid individual/shared configuration	Apply saved arrangement; otherwise Exception Queue
A01	Allocation	Eligible group and reconciled shares are available	Proceed to finalization; otherwise Exception Queue
R01	Rent cycle	Lease/cycle already has a rent charge	Do not generate a second charge

The system will support Individual Billing Arrangement and Shared-Occupancy Billing Arrangement. Under individual billing, the applicable charge will be assigned to the tenant associated with the configured lease or utility account. Under shared



occupancy, the source rent or utility charge will be divided equally among the eligible tenants in the configured billing group. The Manager will identify the property or unit covered by the source charge and maintain the tenant assignments used for allocation. The study will use equal splitting only; it will not estimate consumption or use meter readings to determine individual shares. For shared rent, the eligible tenants will be the active tenants assigned to the billing group on the rent-generation date. For a shared utility charge, the eligible tenants will be those assigned to the billing group on the final day of the source document's billing period. Tenant assignment dates will be used to determine eligibility. A tenant whose assignment has ended before the relevant date will not be included. The eligible tenant list will be recorded with the allocation so that a later change in occupancy does not silently alter a finalized charge. If no eligible tenant can be identified, or if the source period or allocation group cannot be established, the record will be sent to the Exception Queue. The computation will use whole centavos to preserve the source amount. The system will divide the source charge by the number of eligible tenants and assign the same whole-centavo base share to each tenant. Any remaining centavos will be distributed one at a time according to the ascending order of the tenants' permanent record identifiers. For example, a charge of PHP 100.00 divided among three eligible tenants will produce shares of PHP 33.34, PHP 33.33, and PHP 33.33. All displayed shares will use two decimal places, and their total must equal the original source charge before the record can be finalized.

Records placed in the Exception Queue will remain unavailable for finalization until the Manager has reviewed the source information and submitted any required



correction. A correction will identify the affected field, its previous and corrected values, the reason for the change, and the Manager who made it. The corrected record will be revalidated and, when applicable, reprocessed or recomputed. If it still fails a required check, it will remain in the Exception Queue with the unresolved reason recorded. Submission of a correction will not, by itself, indicate successful validation.

Billing Records manages applicable billing information and finalized billing records. The system will use the bill payment statuses Unpaid, Tenant Reported Paid, and Paid. The Tenant may report full payment of the outstanding amount for a selected billing record. Partial payments and overpayments will not be processed automatically within the study. Reporting will change the status to Tenant Reported Paid and record the applicable payment details. The Manager will perform any required verification outside the system before confirming the report. A report that cannot be confirmed may be rejected with a recorded reason, returning the bill to Unpaid and allowing a corrected report. The previous report will remain in the payment history. Only Manager confirmation will record a new payment as Paid and authorize its digital receipt. The confirmed payment will retain its amount, confirmation date, confirming Manager, and unique receipt reference. A repeated request to confirm the same payment will return the existing confirmation and receipt rather than generating another receipt. The system will not independently verify bank transfers, electronic-wallet transactions, cash payments, or direct utility payments. Corrections to finalized billing records will preserve the previous values and their processing history. An unpaid bill may be updated after Manager correction, revalidation, and recomputation. If a payment report is awaiting confirmation

when the billed amount changes, the Manager will review the report against the corrected amount before confirmation. A paid record will not be silently overwritten. Any necessary correction will be recorded as a linked revision showing the original charge, corrected charge, and effect on the recorded balance, while retaining the original confirmed payment and receipt. Adjustments that require an external payment or refund will remain subject to Manager handling and will not introduce automatic settlement. Tenants will be able to view the current billing information through the authenticated portal.

The Centralized Database will retain the relationships among properties, units, tenants, leases, utility mappings, billing sources, allocated charges, payment reports, confirmed payments, receipts, exceptions, corrections, and applicable notification records. A unit under shared occupancy may be associated with more than one tenant through recorded tenant assignments. Each utility allocation will refer to its source bill and eligible tenant group, while each confirmed payment and receipt will refer to the corresponding billing record. These relationships will preserve the source and processing history of a transaction and support the role restrictions described in the study. The architectural framework, flowchart, and DFD present the system's structure and operation; the database design will specify how the corresponding records are stored and related during development.

Tenant billing alerts will use Web Push Notifications as the only external notification channel included in the study. When a tenant grants permission on a

supported browser, the system will create and store the push subscription associated with that tenant's account and device. The implemented notification mechanism will include the background handling required to receive an alert and open the authenticated portal. A notification will direct the tenant to the relevant billing record without treating the alert as payment confirmation or exposing complete billing details before authentication. The system will trigger a billing update when a billing record is finalized or a relevant correction is completed. It will also schedule a reminder three calendar days before the due date using Philippine time. Reminders will apply only to records with an outstanding balance that remain Unpaid; records awaiting Manager confirmation will not receive the ordinary unpaid reminder. If the billing record becomes available after its scheduled reminder time, the system will issue the billing update rather than sending a backdated reminder. A corrected due date will replace any pending reminder schedule. Repeated execution of the same scheduled event will not create duplicate reminder entries. Notification records will distinguish a scheduled alert, a sending attempt, and a failed or accepted request to the push service when those outcomes are available from the implemented mechanism. Acceptance by a push service will not be reported as proof that the tenant received or read the alert. Invalid or expired subscriptions will not prevent billing records from being stored or accessed. Tenants who deny permission, use an unsupported browser, or encounter delivery failure will continue to view their billing details, payment status, and confirmed-payment receipts through the authenticated portal. SMS, GSM, and tenant email notifications will remain outside the scope of the study.

The Admin performs system-level maintenance. The Manager manages property information, tenant records, utility mappings, billing configurations, billing records, exceptions, corrections, payment confirmations, and receipts. The Tenant views billing information and payment status, reports applicable payments, views confirmed-payment receipts, and receives permitted Web Push Notifications. The Utility Provider serves as the source of electronic utility-bill documents received through the designated property-management email account.

The components operate as an integrated web-based property-management system. Property and tenant management, utility mapping, and rent configuration provide the required management information. Billing-processing components support utility-bill intake, OCR extraction, validation, rule-based classification, duplicate checking, billing computation, and exception handling. Billing and payment components manage billing records, payment statuses, and confirmed-payment receipts. The Web Push Notification component supports permitted billing and due-date-related alerts, while the Centralized Database stores the applicable system records. The Admin, Manager, Tenant, and Utility Provider interact with the system according to their defined roles and responsibilities.

2.2 Conceptual Operation

The proposed system operates as an integrated property-management and billing workflow for rent, electricity, and water. Property, unit, tenant, utility-mapping, and rent-configuration information support the processing of rental and utility billing records.

The workflow covers billing intake, OCR extraction, rule-based validation and classification, Exception Queue handling, Billing Computation, Billing Finalization, payment-status management, Digital Receipt generation, tenant portal access, and permitted Web Push Notifications.

Rent is generated according to the tenant's lease-anniversary date using fixed and non-prorated rules. The first billing record includes one-month advance rent as a separate line item. The system then applies the applicable Individual Billing Arrangement or Shared-Occupancy Billing Arrangement before Billing Computation and Billing Finalization. For shared occupancy, applicable rent and utility charges are allocated among active tenants using the predefined rule selected or configured by the Manager. Allocated amounts use two decimal places and must reconcile with the original source billing amount.

Utility billing begins when an electronic utility bill is received through the designated property-management email account. The system retrieves supported standard digitally generated PDF files and sufficiently clear image files, then uses OCR to extract available billing information. The extracted information is checked using predefined, explicit, deterministic, and traceable rules for required fields, Utility Provider and Account/Reference Mapping, billing period, duplicate status, and billing classification. Valid records proceed through the configured billing rules and Billing Computation before Billing Finalization. Incomplete, uncertain, duplicated, unmappable,

or otherwise unprocessable records enter the Exception Queue for Manager correction, followed by revalidation and reprocessing or recomputation when applicable.

A finalized billing record begins with the status Unpaid. A Tenant payment report changes the status to Tenant Reported Paid, but it does not confirm payment. Only the Manager can change the status to Paid, after which the system generates a Digital Receipt. The system may trigger permitted Web Push Notifications for billing updates and due-date alerts, including the configured reminder three days before the due date. These alerts direct Tenants to the authenticated web portal and do not confirm payment or prove notification delivery or reading. SMS and tenant email are not used as notification channels.

Process Flowchart

The process flowchart presents the order of activities and decisions in the billing workflow. It shows the separate rent and utility-bill paths, their convergence at Billing Finalization, the Exception Queue return to validation, payment-status transitions, Digital Receipt generation, and the supporting Web Push Notification branch.

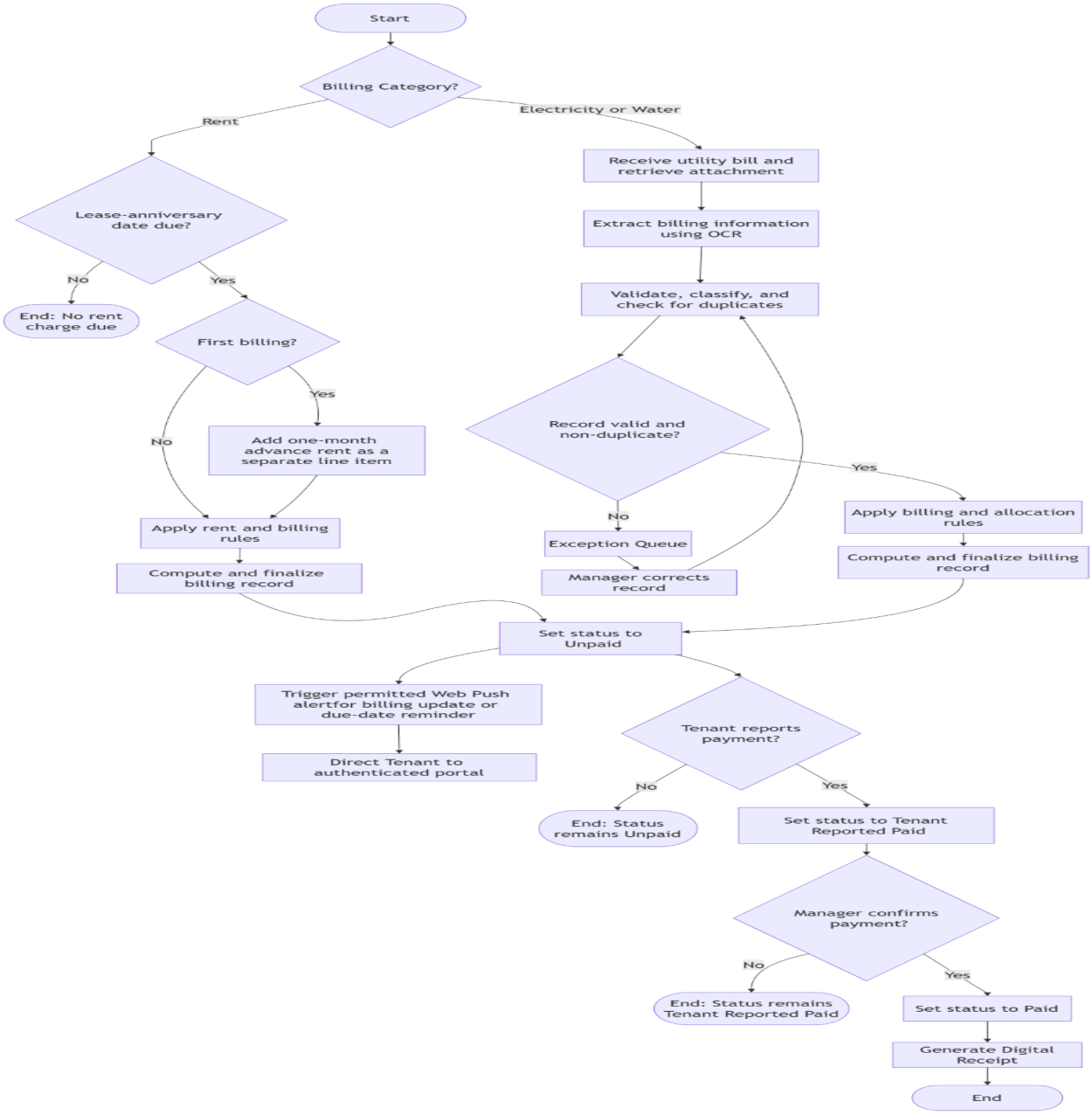


Figure 2.2. Overall Process Flowchart

The flowchart shows the sequence of billing activities and decision points from rent or utility-bill processing through Billing Finalization. It also shows that only Manager

confirmation changes a tenant-reported payment to Paid and permits Digital Receipt generation, while Web Push Notification remains a supporting alert function.

Full-Blown Data Flow Diagram

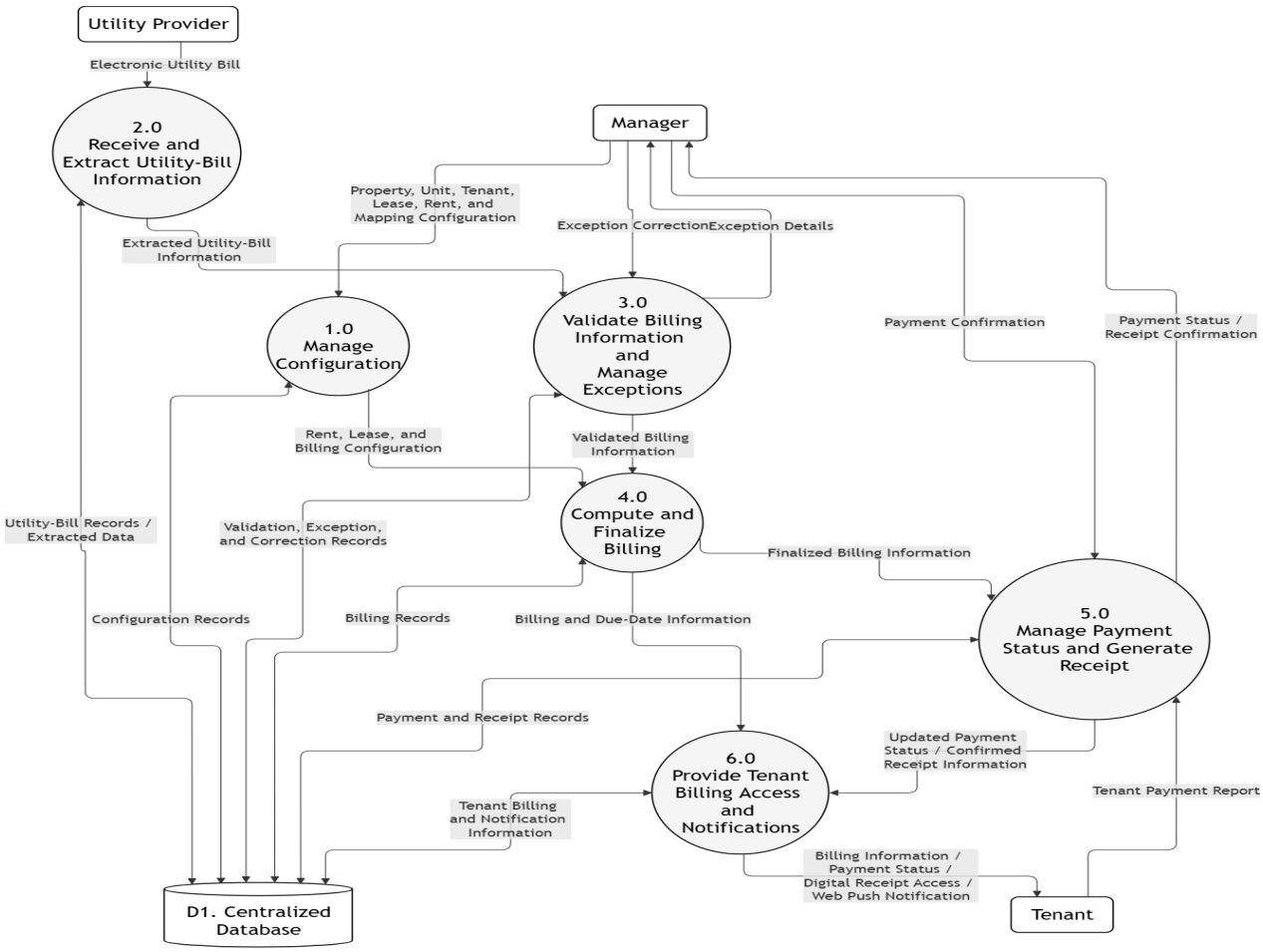


Figure 2.3. Data Flow Diagram

The DFD illustrates how billing-related information moves among the Manager, Tenant, Utility Provider, system processes, and Centralized Database. It shows the flow of configuration data, utility-bill information, billing records, payment information, receipt

information, and tenant notification information through the system processes. External entities interact with the system through the appropriate processes, while the Centralized Database stores and provides the records required by the system.

2.3 Mockup Design

The mockups present the proposed interfaces for the Admin, Manager, and Tenant roles. The Admin interface is limited to the system-maintenance functions included in the study. The Manager interface presents property and tenant records, billing configuration, incoming utility-bill information, the Exception Queue, billing records, and payment confirmation. The Tenant interface presents the assigned billing records, payment-reporting function, confirmed-payment receipts, and applicable notifications. The mockups serve as interface references during development and do not represent completed or evaluated system functions. Their labels, available actions, and displayed payment statuses will follow the approved scope and workflow.

2.4 Methodology of the Study

The study will use the Iterative Software Development Methodology to develop the proposed Intelligent Property Management Workflow with OCR-Enabled Automated Billing Classification and Tenant Notification. This methodology will organize

development into successive iterations in which related system functions will be planned, designed, developed, reviewed, refined, and integrated.

The methodology is appropriate for the proposed system because its functions are interconnected but can be developed in manageable groups. These functions include system foundation, property and billing configuration, utility-bill intake, OCR extraction, rule-based processing, exception handling, Billing Computation, payment and receipt management, tenant access, Centralized Database functions, and Web Push Notifications. The methodology will also support the intended automate by default, intervene by exception approach.

2.4.1 Principles of the Iterative Methodology

The Iterative Software Development Methodology will be guided by five principles. First, incremental development will allow the researchers to develop related system functions in smaller iterations instead of implementing the entire system at once. Second, continuous testing and review will allow each iteration to be checked against the defined requirements, user roles, workflow rules, and data requirements.

Third, refinement between iterations will allow issues identified during development to be corrected before related functions are integrated into succeeding iterations. Fourth, cumulative integration will allow functions developed in earlier iterations to work with functions developed later until they form a complete system.

Finally, exception-based intervention will allow valid records to proceed through the defined rules while records requiring review are directed to the Exception Queue for Manager intervention.

2.4.2 Development Iterations of the Methodology

The proposed system will be developed through seven planned iterations based on the functional requirements of the study. Each iteration will focus on a related group of functions that will later be integrated into the complete system.

Table 2.2: Development Iterations of the Proposed System

Iteration	Development Focus	Main Functions
Iteration 1	Requirements and System Foundation	User roles, authentication, property, unit, tenant, and Centralized Database structure
Iteration 2	Property and Billing Configuration	Utility providers, account or reference mapping, rent configuration, billing arrangements, and billing rules
Iteration 3	Property-Management Email Intake and OCR	Designated email intake, attachment retrieval, supported document processing, and OCR extraction
Iteration 4	Rule-Based Validation and Exception Handling	Required-field checking, mapping, billing-period checking, duplicate checking, billing classification, Exception Queue, Manager correction, and revalidation
Iteration 5	Billing Computation and Finalization	Rent, electricity, and water billing; Individual and Shared-Occupancy Billing Arrangements; allocation; computation; and Billing Finalization

Iteration	Development Focus	Main Functions
Iteration 6	Payment, Digital Receipt, and Web Push Notification	Payment statuses, Tenant payment reporting, Manager confirmation, Digital Receipt generation, tenant portal access, and Web Push alerts
Iteration 7	*System Integration and Testing*	*Integration of system functions, implemented security controls, system-wide testing, and final evaluation*

2.4.3 Application of the Methodology to the Study

The planned iterations will be applied progressively, beginning with the system foundation and configuration functions and continuing with utility-bill intake, OCR extraction, rule-based processing, exception handling, Billing Computation, payment and receipt management, tenant access, and Web Push Notifications. Each iteration will build upon related functions developed in earlier iterations.

During development, each iteration will undergo planning, design, development, testing and review, followed by refinement and integration. Findings from testing and review will be used to correct related functions before succeeding components are integrated. Records requiring Manager intervention will be handled through the Exception Queue according to the defined processing rules.

The final iteration will integrate the developed functions and conduct system-wide testing and evaluation. The completed system will be evaluated using the selected ISO/IEC 25010:2023 quality characteristics: Functional Suitability, Performance Efficiency, Interaction Capability, Reliability, and Security. No completed iteration, testing

result, or evaluation result is claimed in this chapter. ISO/IEC 25010:2023 provides a product-quality model applicable to ICT products and systems.

2.4.4 Benefits of the Methodology to the Study

The Iterative Software Development Methodology will allow the researchers to develop the proposed system in manageable groups of related functions. This will support the early identification and correction of defects and inconsistencies before succeeding functions are integrated.

The methodology will also support continuous refinement as development progresses. Testing and review during each iteration will provide opportunities to improve related functions and maintain consistency among the system requirements, workflows, data structures, and user roles.

Finally, the methodology is appropriate for the proposed system because its functions are interconnected and require gradual integration. The iterative approach will allow the researchers to progressively form the complete system while maintaining the intended automate-by-default, intervene-by-exception approach.

2.4.5 System Testing and Evaluation Procedures

The researchers will test the proposed system during each development iteration and after the system functions have been integrated. Testing will follow the business, user, and security requirements of the study. The test cases will cover ordinary transactions and records that require Manager intervention, including incomplete billing information, unmapped utility accounts, duplicate

documents, and corrected exception records. Each test case will identify the input, the expected result, the observed result, and whether the requirement was satisfied. Defects found during testing will be corrected and the affected cases will be repeated before final evaluation.

OCR testing will compare the extracted amount due, due date, billing period, utility provider, and account or reference identifier with manually checked values from the same supported documents. The researchers will record correct, incorrect, and missing values for each required field. Field accuracy will be computed by dividing the number of correctly extracted values by the number of expected values for that field and multiplying the result by one hundred. Differences in formatting will be accepted only when the declared normalization rules preserve the same amount, date, period, provider, or identifier. The document set and the normalization rules will be established before final testing. Documents used to refine the extraction procedure will be identified separately from those used for final evaluation.

The rule-based workflow will be tested by comparing its validation, mapping, classification, and exception-routing results with the expected outcomes of predefined test cases. Billing computation will be checked against independently calculated rent and utility charges. Shared allocations will be checked both for the correctness of each tenant's share and for agreement between the total allocated amount and the source charge. Payment tests will verify that tenant reporting does not confirm payment and that a digital receipt is generated only after Manager confirmation. Repeated processing of the same record will also be tested to check that the system does not create duplicate billing charges or receipts.

The final evaluation will use the five selected ISO/IEC 25010:2023 quality characteristics: Functional Suitability, Performance Efficiency, Interaction Capability, Reliability, and Security. Functional Suitability will be examined through requirement-based test cases. Performance Efficiency will be examined through measured processing and response times under a stated testing environment. Interaction Capability will be examined through representative user tasks and an adviser-reviewed evaluation questionnaire. Reliability will be examined through repeated processing and recovery tests within the implemented workflow. Security will be examined through authentication, role restrictions, and attempts to access records outside the user's permitted scope. User ratings will supplement the technical findings rather than replace checks of computation, access control, or processing correctness.

Managers and tenants who participate in user evaluation will perform tasks appropriate to their roles. Manager tasks will include reviewing billing information, correcting an exception, and confirming a reported payment. Tenant tasks will include viewing billing details, reporting payment,and viewing a confirmed-payment receipt. The researchers will record task completion and encountered difficulties, then summarize the questionnaire responses using the approved rating scale and scoring procedure. Technical results will be reported through counts, percentages, processing-time summaries, and descriptions of failed cases. The number and selection of evaluators, the document count, and the questionnaire scoring procedure will be established with the adviser before final data collection. No testing or evaluation result is claimed at the proposal stage